

Vishwajeet Jadhav

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Education

Indian Institute of Technology, Roorkee 2022 – Present

B.Tech in Electronics and Communication Engineering

- **CGPA:** 9.582/10.0
- **Department rank:** 2/121
- **Coursework:** Mixed Signal Circuit Design, Analog circuits, Electronic Network Theory, Automatic Control Systems, Signals and Systems, Digital Signal Processing, RF and Mixed Signal Circuits, Fundamentals of Microelectronics, Digital Logic Design, Digital Image Processing, Embedded System Design, Microwave Engineering, Computer Architecture and Organization, Semiconductor physics
- **Other:** Undergraduate Teaching Assistant - ECC-106 (Signals and Systems), Spring 2025

Woodridge High School, Aurangabad 2010-2022

Higher Secondary Education (Class XII) and Secondary Education (Class X)

- **Class XII:** 97.20%
- **Class X:** 96.60%

Research Internship

IUSSTF - Viterbi Research Intern May 2025 – July 2025

University of Southern California, Viterbi School of Engineering

- Research intern at the ACME Lab at USC
- Contributed to the design of a PLL based IC for resonant-shift-based biosensing using magnetic nanoparticles
- Designed PCB for testing and characterization of neurostimulator IC
- Advisor: **Prof. Constantine Sideris**, Associate Professor, ACME Lab, USC Viterbi

Research Experience

Design of Ultra-High Bandwidth Continuous Time Delta Sigma ADC Dec 2024 – Present

Indian Institute of Technology, Roorkee

- Proposed a digital logic based ELD compensation scheme for $ELD > 1$ clock cycles for CT-DSM and verified logic through simulation both in MATLAB and ideal blocks in Cadence
- Targeting a 200MHz bandwidth and 68dB SNDR at sampling rate of 9.6GHz
- Currently designing the modulator in GF 22nm FDSOI technology node
- This project is part of my final year undergraduate thesis project
- Advisor: **Prof. Sarvana Kumar M.**, Assistant Professor, ECE Department, IIT Roorkee

Designing CMOS Potentiostat for Electrochemical Biosensing April 2024 – Jan 2025

Indian Institute of Technology, Roorkee

- Implemented a potentiostat for glucose biosensing in 180nm CMOS technology node in Cadence virtuoso at schematic level
- Have also presented the research at the student research forum at IEEE ISICAS 2024
- Advisor: **Prof. Sarvana Kumar M.**, Assistant Professor, ECE Department, IIT Roorkee

Resistance Modeling in Nanosheet FET's Sept 2023 – May 2024

Indian Institute of Technology, Roorkee

- Worked at the Dirac Lab at IIT-Roorkee in my 2nd year of undergrad
- Modeled variation of extrinsic resistances in nanosheet FETs through device level simulations in Sentaurus TCAD
- Advisor: **Prof. Avirup Dasgupta**, Associate Professor, ECE Department, IIT Roorkee

Semester Projects

Design of a 4th order CT Butterworth Filter in 180nm node in Cadence Virtuoso

ECL-538 : Mixed Signal Circuit Design course

- Implemented the filter as a cascade of two single-ended Tow Thomas Biquad filters
- Each stage of filter utilised a previously designed two-stage Miller compensated OTA, designed using gm/Id technique. Entire design was carried out in SCL 180nm pdk

FRUC detection using periodic properties of edge intensity

ECN-316 : Digital Image Processing course

- Implemented algorithms for detection of video Frame-Rate Up-Conversion (FRUC) using periodic variation of frame edge intensity
- Drop in frame edge intensity was used as a measure to detect upconversion

Academic Achievements

- IUSSTF - Viterbi 2025 fellow. Part of a highly selective 15 student research cohort at University of Southern California for summer of 2025
- Presented my research work on CMOS Potentiostat implementation at the student research forum at ISICAS-2024
- Selected for International Astronomy Olympiad Final Camp Stage (Top 26 nationally)
- KVPY SX Fellow : All India Rank 363
- KVPY SA Fellow : All India Rank 774
- JEE Mains 2022 : All India Rank 156 out of 1M+ candidates
- JEE Advanced 2022 : All India Rank 1369 out of 1M+ candidates
- NTSE (National Talent Search Examination) scholar 2020

Skills

Languages: C/C++ , Verilog HDL, Python, MATLAB

Design and Simulation tools: Cadence virtuoso, MATLAB, KiCAD, Sentaurus TCAD, Xilinx vivado, Ltspice